



Data Science

NVIDIA cuDF Accelerates pandas Nearly 150x with Zero Code Changes



Mar 18, 2024

👍 +60 Like 💬 Discuss (5)

By [Jay Rodge](#), [Nick Becker](#) and [Ashwin Srinath](#)

At NVIDIA GTC 2024, it was announced that [NVIDIA cuDF](#) can now bring GPU acceleration to 9.5 million pandas users without requiring any code changes.



Update: cuDF [now instantly accelerates pandas with zero code changes in Google Colab](#). Try out the [tutorial](#) in a Colab notebook today.



processing speed and efficiency in CPU-only systems.

RAPIDS is an open-source suite of GPU-accelerated Python libraries designed to improve data science and analytics pipelines. RAPIDS cuDF is a GPU DataFrame library that provides a pandas-like API for loading, filtering, and manipulating data. In earlier releases of cuDF, it was meant for GPU-only development workflows.

Last fall, RAPIDS released a version of cuDF that brought accelerated computing to pandas workflows with no code changes through a unified CPU/GPU user experience in open beta. At GTC 2024, NVIDIA announced that the cuDF acceleration of pandas is now generally available in the latest release of RAPIDS v24.02. This feature will be supported by NVIDIA AI Enterprise 5.0 later in the spring.



Accelerate Pandas by Nearly 150X with NVIDIA cuDF

NVIDIA Developer



Watch on

Video 1. Accelerate pandas by nearly 150x with cuDF

In the video, you can see identical pandas workflows running side by side in a Jupyter notebook: one uses pandas with CPU-only, one loads the cudf.pandas extension to accelerate pandas with RAPIDS cuDF.

Bringing a unified CPU/GPU experience to pandas workflows

cuDF has always provided users with top DataFrame library performance using a pandas-like API. However, adopting cuDF has sometimes required workarounds:



heterogeneous hardware.

- Manually switching between cuDF and pandas when interacting with other PyData libraries or tooling specific designed for pandas.

Related posts

In the 24.02 release, cuDF accelerates pandas with zero code changes to address these challenges, in addition to the existing GPU-only experience.

This feature was built for data scientists who want to continue using pandas as data sizes grow into gigabytes and performance slows. When cuDF accelerates pandas, operations will run on the GPU if possible, and on the CPU (using pandas) otherwise. cuDF synchronizes between the GPU and CPU under the hood as needed. This enables a unified CPU/GPU experience bringing best-in-



Get Started with GPU Acceleration for Data Science

```
python -m cudf.pandas script.py
```





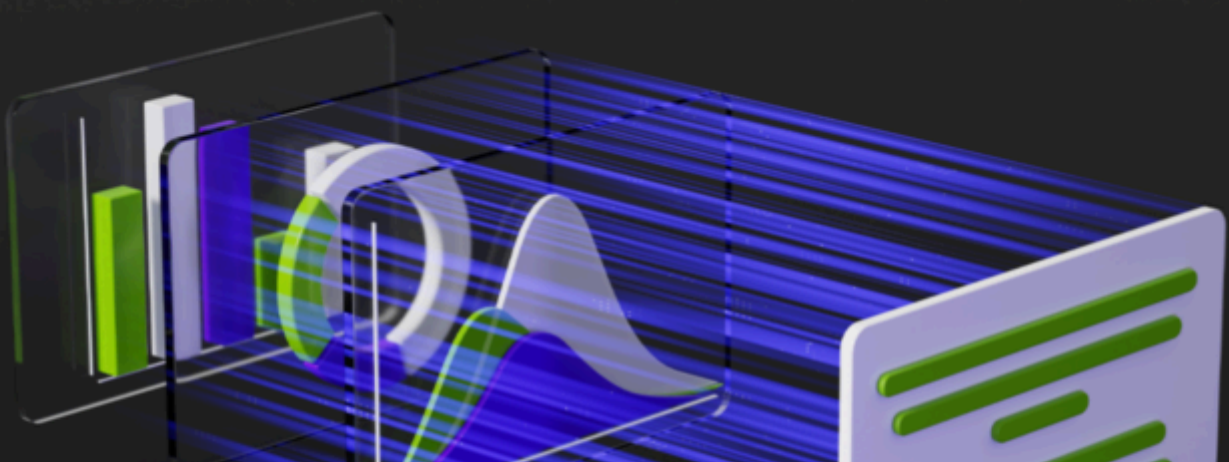
NVIDIA cuDF Unified Memory Accelerates pandas up to 30x on Large Datasets

and working around missing functionality. With cuDF's new pandas accelerator mode, this is no longer an issue. You can run the pandas benchmark code unchanged and achieve significant speedups, using the GPU for most of the operations and the CPU for a small portion to ensure



NVIDIA cuDF Instantly Accelerates pandas up to 50x on Google Colab

HW: NVIDIA Grace Hopper, CPU: Intel Xeon Platinum 8480C | SW: pandas v2.2, RAPIDS cuDF 23.10



Accelerated Data Analytics: Machine Learning with GPU-Accelerated Pandas and Scikit-learn

 Discuss (5)

 +60 Like

Tags

The screenshot shows a Google Colab notebook interface. At the top, the URL bar indicates the notebook is hosted on Google Drive. The notebook title is 'rapids-colab.ipynb'. Below the title bar, there's a 'File Edit View Insert Runtime Tools Help' menu. A button 'OPEN IN PLAYGROUND' is visible. The main content area shows a code cell with the command 'nvidia-smi' executed. The output is a table showing GPU information for a Tesla T4. Below the table, there's a section for 'Processes' showing 'No running processes found'. At the bottom, there's a 'Setup Steps' section with four numbered instructions for installing Miniconda, RAPIDS, and setting environment variables.

Click the Runtime dropdown at the top of the page, then Change Runtime Type and confirm the instance type is GPU.

Then check the output of `nvidia-smi` to confirm you have been allocated a Tesla T4:

```
( ) !nvidia-smi
```

Fri Apr 26 15:38:16 2019

NVIDIA-SMI 418.56		Driver Version: 418.79		CUDA Version: 10.0	
GPU Name	Persistence-M	Bus-Id	Disp.A	Volatile Uncorr. ECC	
Fan Temp Perf	PwrUsage/Cap		Memory-Usage	GPU-Util	Compute M.
0 Tesla T4	Off	80000000:00:04:0	Off		0
N/A 71C P8	18W / 70W	0000:00:00:00	0MiB / 15079MiB	0%	Default

Processes:					CPU Memory
GPU	PID	Type	Process name		Usage
No running processes found					

Setup Steps:

1. Install most recent Miniconda release compatible with Google Colab's Python Install (3.6.7)
2. Install RAPIDS libraries
3. Copy RAPIDS .so files into current working directory, a workaround for conda/colab interactions
4. Set necessary environment variables

```
( ) !wget -O https://repo.continuum.io/miniconda/Miniconda3-4.5.6-Linux-x86_64.sh
!chmod +x Miniconda3-4.5.6-Linux-x86_64.sh
!bash ./Miniconda3-4.5.6-Linux-x86_64.sh -b -f -p /usr/local
!conda install -q -y --prefix /usr/local -c conda-forge \
  -c rapidsai-nightly/label/cuda10.0 -c nvidia/label/cuda10.0 \
  --all
!cp /usr/local/lib/libcudf.so .
!cp /usr/local/lib/librmm.so .

import sys, os
sys.path.append('/usr/local/lib/python3.6/site-packages/')
```

Run RAPIDS on Google Colab—For Free



About Nick Becker

Nick Becker is a senior technical product manager on the RAPIDS team at NVIDIA, where his efforts are focused on building the GPU-accelerated data science ecosystem. Nick has a professional background in technology and government. Prior to NVIDIA, he worked at Enigma Technologies, a data science startup. Before Enigma, he conducted economics research and forecasting at the Federal Reserve Board of Governors, the central bank of the United States.

View all posts by Nick Becker [➤](#)



team developing the CUDA Core Compute Libraries (CCCL) for Python. Previously, he was part of the RAPIDS team at NVIDIA, developing the cuDF Python library. Ashwin is passionate about enabling developers to leverage GPU using high-level programming languages like Python.

View all posts by Ashwin Srinath >

Comments

Notable Replies



November 13, 2023

marcogorelli

Cool!

Did you run the benchmarks using NVIDIA Grace Hopper? How much does it cost? I can't find it



November 13, 2023

benadityan

I see this is only for a Linux environment. Is there a roadmap for cuDf GPU acceleration in standalone Windows (local Jupyter notebook) without WSL.



November 14, 2023

eafpres

I followed the instructions for WSL2, and it installed fine, but in Python it says

```
>>> import cudf.pandas
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ModuleNotFoundError: No module named 'cudf.pandas'
```

Note that the cudf install verifies:

```
eafpres@EAFLLCML:~$ python
Python 3.9.18 (main, Aug 25 2023, 13:20:04)
[GCC 9.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import cudf
>>> print(cudf.Series([1, 2, 3]))
0    1
1    2
2    3
dtype: int64
```



November 16, 2023

todorp

Fantastic performance, when can we expect the Windows version?



March 5, 2024

prisciluis_salas

I am working on a Windows machine, so I tried to do a hard install on a Jupyter notebook, but it failed. Can you help me please?

```
!pip install cudf
```

```
Defaulting to user installation because normal site-packages is not writeable
Collecting cudf
  Using cached cudf-0.6.1.post1.tar.gz (1.1 kB)
  Preparing metadata (setup.py): started
  Preparing metadata (setup.py): finished with status 'done'
Building wheels for collected packages: cudf
  Building wheel for cudf (setup.py): started
  Building wheel for cudf (setup.py): finished with status 'error'
  Running setup.py clean for cudf
Failed to build cudf
```

```
error: subprocess-exited-with-error
```

```
python setup.py bdist_wheel did not run successfully.
exit code: 1
```

```
[47 lines of output]
```

```
running bdist_wheel
```

```
running build
```

```
C:\ProgramData\anaconda3\Lib\site-packages\setuptools\distutils\cmd.py:66: SetuptoolsDeprecationWarning: setup.py install is deprecated.
```

```
!!
```

```
*****
```

```
Please avoid running ``setup.py`` directly.
```

```
Instead, use pypa/build, pypa/installer or other
standards-based tools.
```

```
See https://blog.ganssle.io/articles/2021/10/setup-py-deprecated.html for details.
```

```
*****
```

Continue the discussion at forums.developer.nvidia.com

Participants





DEVELOPER



Join



DEVELOPER

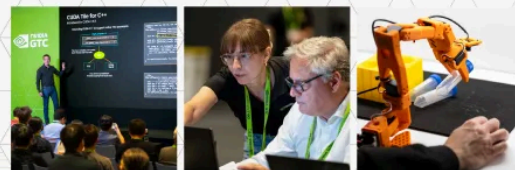


Join



Learn. Build. Get Certified.

Experience hands-on workshops, training labs, and certification at GTC Berlin.

[Learn More](#)

Sign up for NVIDIA News

[Subscribe](#)

Follow NVIDIA Developer

